Ques: Find Median in Data Stream

1, 2, 4, 3, 8, 6, 4, 2, 7, 6

, , , , , 6, 4, 2, 7, 6

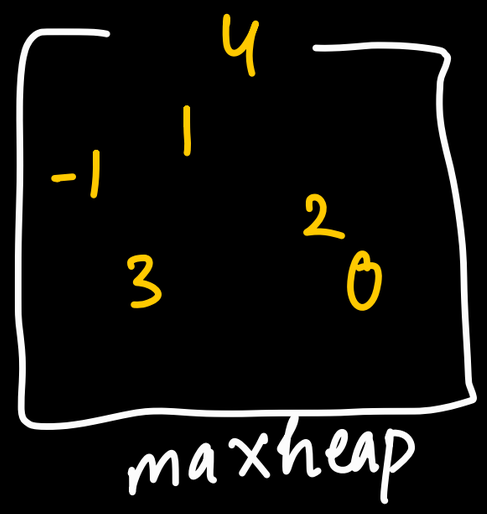
$$T.C. = \sum_{r=1}^{n} r \log r$$

$$\approx n^2 \log n$$

elements		median
1	1	1
1 2	1 2	1.5
1 2 4	1 2 4	2
1 2 4 3	1 2 3 4	2.5
1 2 4 3 8	1 2 3 4 8	3

Ques: Find Median in Data Stream

6 4 3 1 2 5 7 8 9 10 0 -1

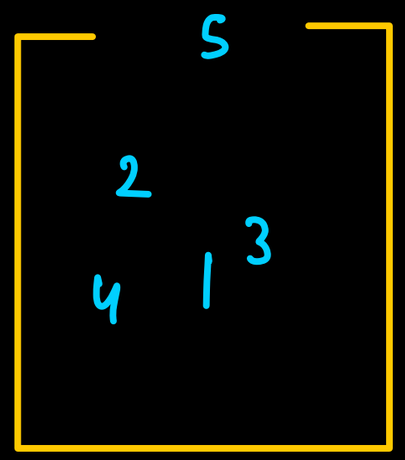


getmedian()

↓

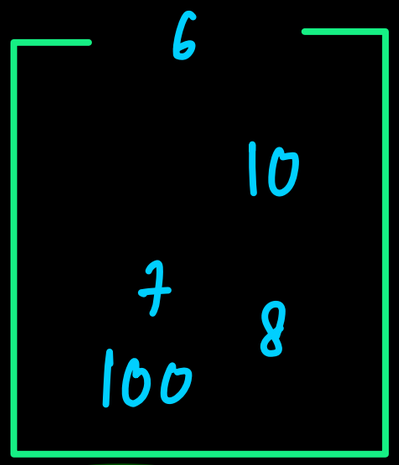
5

Ques: Find Median in Data Stream



maxheap

will contain
n/2 smallest



minheap

will contain
n/2 largest

heap + greedy

Ques: Smallest Range covering elements from K Lists

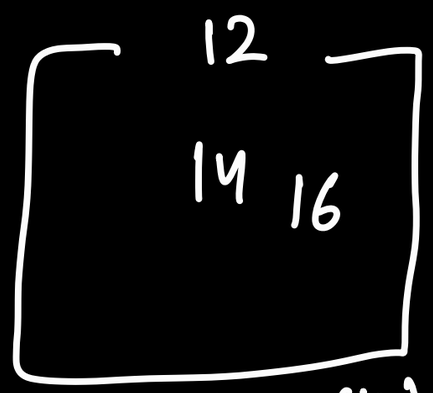
range = [a, b]

↓

minimise 'b-a'

we cannot decrease b

arr = { {4, 7, 9, 12, 15},
{0, 8, 10, 14, 20},
{6, 12, 16, 30, 50} }



minheap(k)

max = ~~6~~ 8 ~~12~~ ~~14~~ 16

smallest range

~~[0, 6]~~
~~[4, 8]~~
[6, 8]

[0, 6], [4, 8], [6, 8], [7, 12], [8, 12], [9, 12], [10, 12]
[12, 14], [12, 16]